NEWCOMER CREDIT DECISIONING COPILOT

Deciding, and defending, thin-file credit for UAE newcomers

A B2B copilot that gives a UAE credit officer a fast, consistent, defensible **approve, decline, or refer** decision on an applicant with no local credit history.

PMCurve AI-PM Capstone, Cohort 6 | Solo, bring-your-own project | Synthetic data only

THE PROBLEM

A new arrival starts with a blank credit file

- **The market.** The UAE is about 88% expat, so a large share of loan applicants are recent arrivals.
- **The gap.** A newcomer has no record at the Al Etihad Credit Bureau (AECB), so the usual credit score does not exist.
- **Today.** Banks reject good customers outright, or assess them slowly and inconsistently by hand: the better part of an hour per file.
- **The cost.** Lost good customers, slow onboarding, and decisions an officer cannot easily defend to a regulator.

88% expat is a market figure. All applicant data in this deck is synthetic.

THE LANDSCAPE

Five ways a bank solves this today, and the gap each leaves

Player	What it represents	The gap it leaves
AECB × Nova Credit Passport	Import home-country bureau history	Only covers applicants from networked countries with a usable home file
Newgen Agentic Decisioning	An explainable decisioning engine	Generic engine, not newcomer-specific alt-data scoring
Liv (Emirates NBD)	The incumbent digital bank app	A consumer brand, no explainable thin-file underwriting
Tabby / Tamara (BNPL)	Proof alt-data underwriting works	Small short-tenor, opaque models, not regulator-grade
CRIF Lending Journey	A platform to license vs build	A blank toolkit; the bank still designs the newcomer logic

None owns the whole: newcomer alt-data scoring, a UAE policy check, and a defensible decision for the applicants the others leave invisible. That gap is the product.

WHO IT IS FOR

The credit officer, who has to stand behind the call

- **User.** A credit officer at a UAE digital bank, assessing newcomer personal-loan and credit-card applications.
- **Their job.** Decide repay-vs-default risk, apply lending policy, and write a rationale they can defend later.
- **Their constraint.** In a regulated decision, being able to explain and reconstruct the call matters as much as the call itself.
- **The design rule.** The tool recommends; the human always makes the final decision.

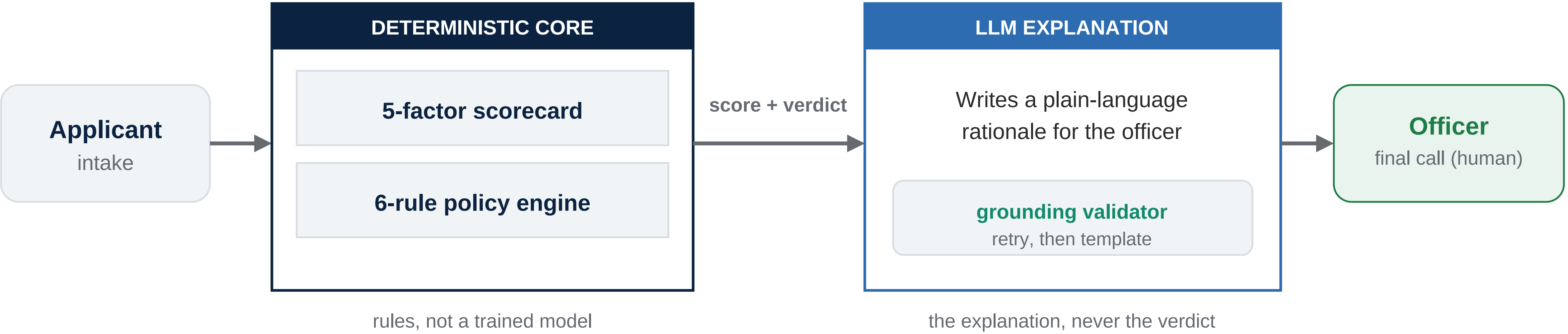
THE APPROACH

Rules plus an LLM, with no trained model

- **What we built.** A rules-based scorecard and policy engine for the decision, with an LLM for the explanation layer only.
- **What we did not build.** A trained credit model. No black-box score drives the verdict.
- **Why.** For a regulated v1 lending decision, explainability and defensibility beat a marginal accuracy gain from a model nobody can audit.
- **The featured layer.** The explanation and decision-reasoning, the part a bank cannot get from a raw prompt.

HOW IT FITS TOGETHER

Deterministic decision, LLM explanation, human gate



The score and the verdict are fully deterministic (approve, decline, or refer). The LLM never touches them: it writes the explanation only, and the officer makes the final call.

THE PRODUCT

Structured intake to a defensible recommendation

1
Intake
Employment, salary, tenure, rent history, and a few alt-data fields.

2
Scorecard
A transparent 5-factor judgmental scorecard scores repay-vs-default risk.

3
Policy check
Six UAE lending rules run, each with a citation.

4
Explanation
An LLM writes a plain-language rationale grounded only in the above.

5
Recommendation
Approve, decline, or refer. The officer makes the final call.

The score and the verdict are fully deterministic. The LLM writes the explanation only; it cannot change points, rules, or the recommendation.

NOT A CONCEPT, A WORKING PRODUCT

A seven-screen officer console, built in Next.js

1 **Assess an applicant**
structured intake to a decision: score, rule checks, explanation

2 **Review queue**
referred cases an officer works; the action persists across a reload

3 **Audit log**
every decision recorded and reconstructable months later

4 **Policy**
the live UAE ruleset, read-only, each rule with its citation

5 **Policy versions**
version history, with each decision stamped by ruleset version

6 **Policy impact**
champion vs challenger on the 24 profiles, with a live what-if

7 **Overview**
the dashboard that ties the workflow together

Type-clean and tested. In v1 the decision logic and logs run in the browser; no applicant data leaves the machine. The score and verdict run without any model call.

A REAL DECISION

What the officer sees, end to end

INPUT	OUTPUT: APPROVE | low risk | 98/100
Government employee, 14 months in the UAE	✓ Salary over double the AED 8,000 minimum; debt burden 18% (cap 50%)
Salary AED 20,000 / month, no obligations	✓ Government employer, top stability tier
On-time rent, 6+ months	✓ All six policy rules pass, each cited
Personal loan: AED 40,000 over 12 months	✓ 6+ months on-time rent: strongest behavioral signal

A weaker profile returns **refer** for a human to check, or **decline** with the exact rule that failed named. Real output, captured from the running app.

THE AI INTEGRATION

An explanation you can trust, by construction

- **The use case.** The LLM turns the deterministic score and policy results into a plain-language rationale for the officer.
- **Grounding validator.** A check catches any invented fact, retries once, and falls back to a deterministic template, so no ungrounded text reaches the officer.
- **Result.** Hallucination is enforced to zero, not just measured. In the eval, 24 of 24 explanations were grounded on the first attempt.
- **Adversarial probe.** A built-in prompt-injection test proves a malicious input in the name field cannot move the decision.

EVALUATION, THE TEST SET

An exam the model could not study for

- **24 locked profiles.** Synthetic applicants spanning every rule and boundary: 8 approve, 8 decline, 8 refer.
- **Targets set in Phase 1.** The metrics below were our success criteria, defined before the build, not chosen to flatter the result.
- **Labels locked first.** Outcomes were fixed before any scorecard weight existed, so the labels could not be tuned to fit the model.
- **Honesty rule.** Boundary mismatches are surfaced, not tuned away. A mismatch is information about a borderline case, not a bug to hide.

EVALUATION, THE RESULTS

Every safety target met; one latency tail to fix

Metric	Target (set in Phase 1)	Measured	Met
Decision accuracy	80% or higher	20/24 (83.3%)	Yes
False approval rate	under 10%	0/24 (0.0%)	Yes
Hallucination (shown)	0 across the set	0/24	Yes
Policy grounding	100% traceable	100%	Yes
Latency per assessment	under 15s	avg 8.2s, max 15.3s	Tail

Creditworthy newcomers approved: 7 of 8. The four accuracy misses are conservative refers or a cautious call; none is a false approval. The only miss vs target is the latency tail, one slow case at 15.3s.

STANDING BEHIND THE DECISION

Each risk has a guardrail and a test, not a hope

The risk	How it is contained and tested
A hallucinated fact in the explanation	Grounding validator catches it, retries, falls back to a template; 0/24 ungrounded, plus an injection probe
A wrong or unsafe approval	24-profile locked evals; 0 false approvals; borderline cases refer, never auto-approve
A decision nobody can defend later	A policy citation on every rule, plus the ruleset version stamped on every decision, in an audit log
A bad policy change shipping	Champion vs challenger run on all 24 profiles before any rule change goes live

The point: a prompt proves the model works once. Production needs evals, override tracking, and decision reconstruction. That is what a bank cannot get from a raw LLM.

THE ARCHITECTURE BET

Policy is configuration, a market is a pack

- **The rules live in a config pack,** not in code: rule values, citations, score tiers, and band cut-offs are all data.
- **Live what-if.** Drag the debt-burden cap from 50% to 45% and the 24-profile run recomputes on stage, with no code change.
- **The shift is visible.** Approvals drop from 7 of 8 to 6 of 8, two boundary cases move to decline, and every citation now reads 45%.
- **The payoff.** A second market, a new bureau and regulator, is a new pack, not a new project.

THE IMPACT

From the better part of an hour to minutes

TODAY | manual review

- Assemble salary, visa, obligations, bureau status by hand
- Check each against policy from memory or a PDF
- Write up the rationale
- The better part of an hour per file

WITH THE COPILOT

- ✓ Structured intake
- ✓ Policy rules run instantly, with citations
- ✓ Officer reviews the recommendation and counterfactuals
- ✓ Judgment spent only on the refer queue: minutes

Illustrative, showing where the time goes; not a measured timing claim. A measured study is parked for v2.

WHAT IS NEXT

Parked on purpose, ready to answer 'and then?'

- **A second market pack.** Saudi Arabia, swapping AECB and CBUAE for SIMAH and SAMA. The real test of 'a market is a pack'. (Rule values illustrative until SAMA-sourced.)
- **Overrides as feedback.** Every queue override is labeled data on where the rules disagree with human judgment, feeding rule tuning.
- **Version vs version.** A second ruleset (uae v1.1) and a version selector, so impact analysis becomes champion vs challenger.
- **Close the latency tail.** The one item still short of target: isolate the slow case and cap it.

Built, evaluated, and defensible

- All four capstone phases complete: scope, design, build, and evaluation.
- The MVP is built, type-clean, tested, and measured against a locked baseline.
- Every safety target met: 0 false approvals, 0 hallucinations, 100% grounding.
- Live deployment is the one remaining step.

github.com/Mxjoshi/newcomer-credit-copilot

Product, scope, and sign-offs: Monika.   Engineering pair: Claude Code.